UD6
Activity 2

David Nagy György

1r DAM DB

Angela Maria Solarte Bolaños

ÍNDICE

Objective **3**
Description **3**
Part A 3
Part B 4
Part C 6
Part D 7
Activity Submission **8**

Objective

Practice the use of aggregation functions (COUNT, SUM, AVG, MIN, MAX) to obtain summarized information from a normalized cinema database.

Description

Using the cinema database, write SQL queries to answer the following questions.

Before writing the queries, insert sample data into the database. Each table should contain at least 5 to 7 records in order to properly visualize the results of the JOIN operations.

Part A

1. **Total number of screenings**
   Calculate the total number of screenings in the database.

```
SELECT COUNT(*) AS 'Total_Projections' FROM Projection;
```

Filter Rows: Export: Wrap Cell Content:

#	Total_Projections
	5

Result 3

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:29:53	SELECT COUNT(*) AS 'Total_Projections' FROM ...	1 row(s) returned	0,00095 sec / 0,...

2. **Screenings per movie**
   List each movie title with its total number of screenings.

```
    -- 2
SELECT m.title, COUNT(*) FROM Projection p
INNER JOIN Movie m ON p.id_movie = m.id_movie;
```

Filter Rows: Export: Wrap Cell Content:

#	title	COUNT(*)
	Viernes 13	5

Result 4

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:32:53	SELECT m.title, COUNT(*) FROM Projection p IN...	1 row(s) returned	0,0043 sec / 0,0...

3. **Screenings per cinema**

List each cinema with the total number of screenings held there.

```
SELECT c.cine_name, COUNT(*) AS 'Total_Projections' FROM Projection p
INNER JOIN Cine c ON p.id_cine = c.id_cine;
```

Filter Rows: Export: Wrap Cell Content:

#	cine_name	Total_Projections
	Cine Centro	5

Result 5

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:39:14	SELECT c.cine_name, COUNT(*) AS 'Total_Proje...	1 row(s) returned	0,0035 sec / 0,0...

Part B

4. **Screenings in a specific cinema**

Count the total number of screenings in the cinema *"Cine Centro"*.

```
SELECT c.cine_name, COUNT(*) AS 'Total_Projections'
FROM Projection p INNER JOIN Cine c
ON p.id_cine = c.id_cine
WHERE c.cine_name = 'Cine Centro';
```

Filter Rows: Export: Wrap Cell Content:

#	cine_name	Total_Projections
	Cine Centro	1

Result 6

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:49:08	SELECT c.cine_name, COUNT(*) AS 'Total_Proje...	1 row(s) returned	0,0012 sec / 0,0...

5. **Screenings by director**

List each director with the total number of screenings of their movies.

```
SELECT d.director_name, COUNT(*) AS 'Total_Projections', m.title
FROM Projection p INNER JOIN Director d
ON p.id_director = d.id_director
INNER JOIN Movie m ON p.id_movie = m.id_movie
GROUP BY d.director_name;
```

Filter Rows: Export: Wrap Cell Content:

#	director_name	Total_Projections	title
1	Santiago Segura	1	Viernes 13
2	William Shakespeare	1	Torrente
3	Marlon	1	NULL
4	Dandy	1	NULL
5	Emilio	1	Titanic

Result 8

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:59:18	SELECT d.director_name, COUNT(*) AS 'Total_Pr...	5 row(s) returned	0,0013 sec / 0,0...

6. **Screenings by genre**

List each genre with the total number of screenings, ordered from highest to lowest.

```
SELECT m.genre, COUNT(*) AS 'Total_Projections'
FROM Projection p INNER JOIN Movie m
ON p.id_movie = m.id_movie
GROUP BY m.genre
ORDER BY 'Total_Projections' desc;
```

Filter Rows: Export: Wrap Cell Content:

#	genre	Total_Projections
1	NULL	2
2	Terror	1
3	Comedia	1
4	Drama	1

Result 9

Action Output

#	Time	Action	Message	Duration / Fetch
1	18:12:02	SELECT m.genre, COUNT(*) AS 'Total_Projection...	4 row(s) returned	0,0011 sec / 0,0...

Part C

7. **Movie with the most screenings**

Identify the movie with the highest number of screenings.

```
SELECT m.title, COUNT(*) AS 'Total_Projections'
FROM Projection p INNER JOIN Movie m
ON p.id_movie = m.id_movie
GROUP BY m.title
ORDER BY 'Total_Projections' desc
LIMIT 1;
```

Filter Rows: Export: Wrap Cell Content: Fetch rows:

#	title	Total_Projections
	Viernes 13	1

Result 1

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:12:28	SELECT m.title, COUNT(*) AS 'Total_Projections' ...	1 row(s) returned	0,0078 sec / 0,0...

8. **Top 3 cinemas with the most screenings**

Show the three cinemas with the highest number of screenings.

```
SELECT c.cine_name, COUNT(*) AS 'Total_Projections'
FROM Projection p INNER JOIN Cine c
ON p.id_cine = c.id_cine
GROUP BY c.cine_name
ORDER BY 'Total_Projections' DESC
LIMIT 3;
```

Filter Rows: Export: Wrap Cell Content: Fetch rows:

#	cine_name	Total_Projections
	Cine Centro	1
	Cine Grande	1
	Ocimax	1

Result 2

Action Output

#	Time	Action	Message	Duration / Fetch
1	19:18:21	SELECT c.cine_name, COUNT(*) AS 'Total_Proje...	3 row(s) returned	0,0024 sec / 0,0...

9. **First and last screening per movie**
   For each movie, show the date and time of its first and last screening.

```
    -- 9
SELECT m.title, MAX(p.projection_Datetime) AS 'Last_Screenings', MIN(p.projection_Datetime) AS 'First_Screenings'
FROM Movie m INNER JOIN Projection p ON m.id_movie = p.id_movie
GROUP BY 'Last_screenings', 'First_Screenings'
LIMIT 2;
```

Filter Rows: Export: Wrap Cell Content:

#	title	Last_Screenings	First_Screenings
1	Viernes 13	2025-04-01 22:00:00	2024-01-01 19:00:00

Result 2

Action Output

#	Time	Action	Message	Duration / Fetch
1	17:49:...	SELECT m.title, MAX(p.projection_Datetim...	1 row(s) returned	0,00029 sec /...

Part D

10. **Most screened movie in "Cine Centro"**
    Find the movie with the highest number of screenings in the cinema *"Cine Centro"*.

```
-- 10
SELECT p.projection_ID, COUNT(p.projection_ID) AS 'number_Sreenings', c.cine_name
FROM Projection p INNER JOIN Cine c ON p.id_cine = c.id_cine
WHERE c.cine_name = 'Cine Centro'
GROUP BY 'number_Screenings'
LIMIT 1;
```

Filter Rows: Export: Wrap Cell Content: Fetch rows:

#	projection_ID	number_Sreening	cine_name
1	1	1	Cine Centro

Result 1

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:56:...	SELECT p.projection_ID, COUNT(p.projecti...	1 row(s) returned	0,0024 sec / ...

11. **Analysis of screenings by cinema and genre**
    For each cinema whose name starts with *"Cine"* or whose address contains *"Centro"*, show the genre, the total number of screenings, and the date and time of the first and last screening, only for genres containing *"Acc"* or *"Dra"*, and excluding movies whose title contains *"Saga"*.

```sql
-- 11
SELECT
    c.cine_name,
    c.cine_address,
    m.genre,
    COUNT(p.projection_ID) AS total_projections,
    MAX(p.projection_Datetime) AS Last_Screenings,
    MIN(p.projection_Datetime) AS First_Screenings
FROM Projection p
INNER JOIN Cine c ON p.id_cine = c.id_cine
INNER JOIN Movie m ON p.id_movie = m.id_movie
WHERE
    (c.cine_address LIKE '%Cine%' OR c.cine_address LIKE '%Centro%')
    AND (m.genre LIKE '%Acc%' OR m.genre LIKE '%Dra%')
    AND m.genre NOT LIKE '%Saga%'
GROUP BY
    c.cine_name,
    c.cine_address,
    m.genre
ORDER BY
    total_projections DESC,
    Last_Screenings DESC,
    First_Screenings DESC;
```

Filter Rows: Export: Wrap Cell Content:

#	cine_name	cine_address	genre	total_projection	Last_Screening	First_Screening

Result 2

Action Output

#	Time	Action	Message
1	13:33:...	SELECT c.cine_name, c.cine_address, m.genre, COUNT(p.projection_ID) AS total_projections, MAX(p.projection_Datetime) AS Last_Screenings, MIN(p.proje...	0 row(s) returned

It's nothing screening because en my DB I don't have this genre's.

Activity Submission

Export your answers to PDF format and submit them via Google Classroom.